Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First / Second Semester 2023-2024

Mid-Semester Test
(EC-2 Regular)

Course No. : DSECLZG529/AIMLCZG529
Course Title : Data Management for Machine Learning
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 20/7/2024 (AN)

No. of Pages = 2
No. of Questions = 5

Note to Students:
1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

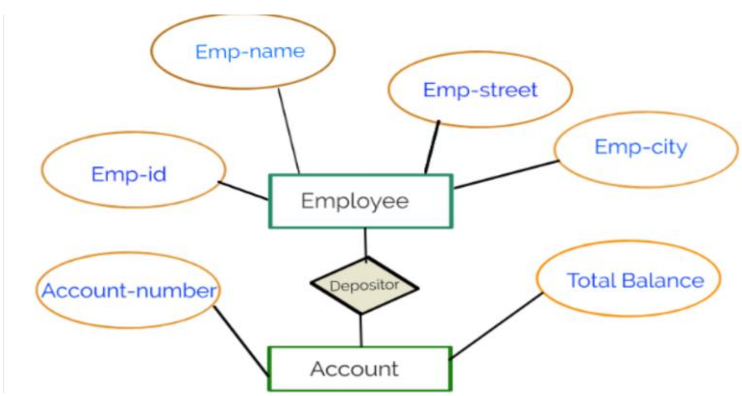
Q.1(A) What file format should program data be stored in (example: .JSON, .csv, .txt, etc.)? **[3 Marks]**

Q.1(B) Metropolis Pathology Lab, Delhi has provided the data as further. As per the doctors prediction, the150 persons are healthy and 350 of them are diabetic. But the pathologist discover that there are indeed 305 persons are diabetic and there are only 195 ones are healthy. As per the pathologists the test data is as follows :

		Predicted Data	
		Healthy	Diabetic
Actual Data	Healthy	45	150
	Diabetic	105	200

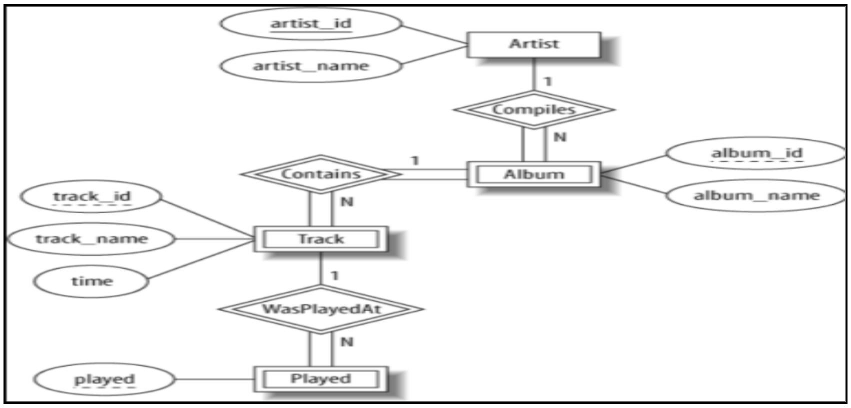
Calculate accuracy of the disease prediction by the doctors **[3 Marks]**

Q.2(A) Consider the following Entity-Relationship Diagram representing Employee details
Represent this ERD in the following data format using sample data
i) Row-major format ii) Columnar format [2 + 2 = 4 marks]



Q.2(B) Consider the following Entity-Relationship Diagram representing Online Radio System .
Represent this ERD in the following data format using sample data.

i) CSV ii) JSON [2 + 2 = 4 marks]



Q.3 You are a data scientist at a healthcare company working on a project to predict patient readmissions within 30 days of discharge. The project involves handling large volumes of patient data from electronic health records (EHR), developing a machine learning model, and deploying the model into production. The workflow involves a Data Pipeline and a Machine Learning Pipeline. Explain how you would set up and manage the **Data Pipeline** and **Machine Learning Pipeline** for this project. Include the activities performed at each stage and justify the importance of each component in the overall workflow.

[3 +3 = 6 Marks]

Q4) Assume you are developing a Travel Booking app that allows users to search for Flights, Hotels, Vacation Rentals, Car Rentals, and Travel Packages for booking. This app aims to fulfill all travel needs for planning, booking, and managing trips. Whether it's a family vacation, a business trip, or a weekend getaway, the app should feature listings from top travel agencies, hotels, and airlines, as well as user-generated reviews and no-commission options. This travel search app should make trip planning on mobile simpler, faster, and more efficient. When users are online searching for travel options, the app should provide recommendations for relevant travel services and display targeted advertisements to enhance the user experience and increase ad conversion rates.

a) Identify any two approaches through which this prepared report can be served to the travel agency or its representative.

b) Assume a travel agency is promoting its travel packages on this application and needs a periodic report on searches for travel packages, type of information accessed, brochure downloads, queries raised against packages, etc. What type of data abstraction will be required to generate such a report? Why?

c) Discuss briefly the data flow (and the stages involved) that will be required to meet this requirement along with a pictorial representation.

d) Who are the primary users of this system and what type of user interfaces are suitable for them to interact with this application? **[1+1+2+2 = 6 Marks]**

Q5) XYZ is developing a student management system that needs to manage diverse data, such as student profiles, course enrollments, grades, and extracurricular activities. Considering the need for flexibility in data structure and handling complex, hierarchical data, why might XYZ select a document model for their database? Provide an example of how the data might be organized and include a **sample code snippet** for the student data structure

[4 Marks]

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